

Engineering analysis in imprecise geometric models

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ABSTRACT

Holes, gaps, dangling boundaries and other imperfections of the geometric models preclude direct application of traditional engineering analysis tools. In such cases geometric inaccuracies have to be removed using a geometry “healing” (repair) procedure which results in a valid solid model. Repair procedure applied to the geometric model is computationally expensive and often requires human intervention and supervision. On the other hand, the repair procedure applied to the surface meshes derived from the boundaries of a geometric model may negatively affect the quality of the Finite Element mesh whose construction follows the repair procedure.

In this paper we describe a novel numerical technique that enables engineering analysis in imprecise geometric models without reconstructing a valid solid model. At the heart of the proposed method lies a modified geometrically adaptive integration technique. It uses a hybrid geometric model, that consists of a hierarchical space decomposition, boundary representation (B-rep) and distance fields. Hierarchical space decomposition helps to resolve the geometric imperfections, while the original geometric model is used to allocate the integration points in the boundary (geometry) cells. The proposed method uses solution structures that combine together the distance fields to the geometric boundaries, boundary conditions and basis functions to enforce the prescribed boundary conditions.

Our approach has been verified on several numerical examples. Our numerical experiments confirm high reliability of the proposed engineering analysis approach for a wide range of geometric imperfections. Despite that the paper presents 2D examples the proposed approach can be generalized in 3D.

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1. Introduction

Engineering analysis is one of the important components of every design process. It accelerates product development, ensures safety and durability of the products. Currently, life of almost every product starts from creation of a Computer Aided Design (CAD) geometric model. During design-analysis cycle geometric models are modified to accommodate the changes dictated by the analysis results, shape optimization, etc. In many cases extensive editing of CAD models and their translation from one CAD system to another result in geometric inaccuracies and inconsistencies [1]. There are two major sources of errors in geometric models: inconsistencies and inaccuracies presented in the geometric representation; and geometric errors due to conversion from one CAD system to another [2]. Inconsistencies and inaccuracies in a geometric representation such as, for example, free edges (edges connected to one face), non-manifold boundaries, sliver faces and inaccurately computed intersections of the surface patches are

accumulated during editing of geometric models. Some of these geometric imperfections are shown in Fig. 1(a). Acquired and reverse engineered geometric models which are represented as a “polygonal soup” [3] may contain holes in their boundaries due to the missing (non-scanned) pieces (Fig. 1(b)). Incorrectly chosen tolerances in the stereolithography (STL) geometric models [4] may produce either gaps or non-manifold boundaries. Conversion of the geometric models from one CAD system to another may result in loss of semantics information and numerical precision. Because different CAD systems use different geometric tolerances and geometric algorithms, loss of numerical precision often causes appearance of gaps, misaligned boundaries, and topological inconsistencies in geometric model which undefines the notion of “inside” and “outside”.

Despite that these geometric imperfections can be very small and almost invisible, they often prevent Finite Element (FE) meshing and direct application of engineering analysis methods based on the Finite Element paradigm. The need to perform the analysis in the presence of geometric inaccuracies and imperfections led to the development of various geometry healing (repair) techniques and tools [5] that can be classified as the volumetric [6], surface [7–11] and hybrid [12] healing techniques. Volumetric techniques [6] represent the surface of the geometric model in the

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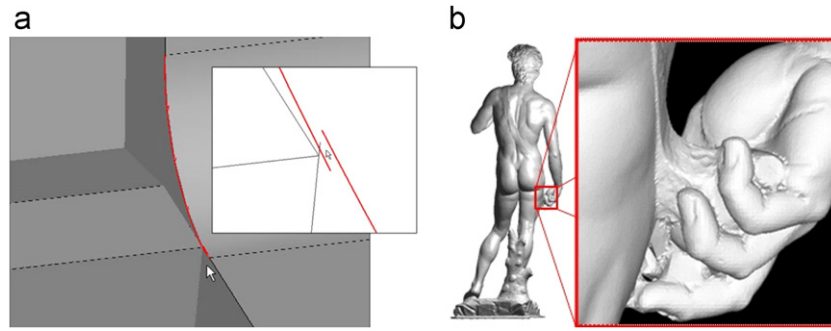


Fig. 1. (a) Inconsistencies and inaccuracies in a geometric representation such as, for example, free edges (edges connected to one face), non-manifold boundaries, and inaccurately computed intersections of the surface patches are accumulated during editing of geometric models. (b) Noisy triangulated surface of the Michelangelo's David statue contains holes due to the limitations of the laser scanning technology.

volumetric domain and apply the healing methods to the volumetric model. After fixing geometric imperfections, the volumetric model is transformed back to the surface model using either marching cubes or dual contouring algorithms [13,14]. Application of volumetric healing methods produces manifold boundaries, but these methods often result in over tessellated geometric boundaries. Surface-based geometry repair techniques perform local modifications of the original model at the places where the geometric errors and inconsistencies have been detected. These techniques are directly applied to the tessellated (usually triangulated) boundary [8]. A surface repair method applied directly on a Boundary representation (B-rep) geometric model was proposed in [9]. Geometric computations are used to determine intersections of the neighboring surface patches, projecting and inserting boundary edges into geometric faces [12]. Gaps between tessellated surface patches are removed by using stitching [10] or zippering [15] algorithms. Surface healing methods often generate small or skewed triangles that lead to poor FE meshes. Despite that automated geometry repair tools [16] have been proposed, in many real-world cases they usually require human supervision and intervention [17]. Geometry repair also requires careful analysis of a variety of special cases of possible geometric defects and the ways how they can be removed [18,19,11]. Recently, knowledge based and learning based geometry healing algorithms were proposed in [20,21]. An adaptive Cartesian mesh generation approach was proposed in [17]. It is based on volumetric geometry healing and works well for small geometric imperfections.

In this paper we propose and demonstrate a novel numerical technique that enables engineering analysis in imprecise geometric models without reconstructing a valid geometric model. It is based on the solution structure method [22–24] and modified geometrically adaptive integration technique which will be presented in Section 3.2. The proposed engineering analysis method is using the boundaries of the *original* geometric model to enforce the prescribed boundary conditions. It is also used to place the integration nodes. The proposed method requires computations of a Point Membership Classification (PMC) at the nodes of hierarchical space decomposition. These PMC values are then used by a geometrically adaptive integration [25,23] to place integration or collocation points at which the governing equation is enforced. For the valid geometric models PMC can be computed by ray casting/stabbing [6] or by computing the sign of a signed distance to the boundaries of a geometric model, etc. However, imprecisions in the geometric model prohibit direct computation of PMC. Also, as we pointed out earlier, they make the notion of “inside” and “outside” undefined. To compute PMC values for imprecise geometric models we adopt an approach similar to the one described in [6,26,5]. The main difference between the proposed technique and traditional

approaches lies in the fact that our method involves the repair of neither geometrically conforming meshes nor the original geometric model. Instead, it uses a composite geometric representation to resolve geometric inaccuracies and allocate the integration points without reconstructing a valid solid model.

Besides the governing equation, the solution method has to provide the means of satisfying the prescribed boundary conditions. The salient feature of the solution structure method is the exact treatment of the specified boundary conditions by using *solution structures*—expressions that combine boundary conditions, basis functions and the functions vanishing on the geometric boundaries. The method is essentially meshfree and does not require construction of spatial meshes that conform to the shape of a geometric model. Also, it does not restrict the choice of the basis functions used for solution approximation or the solution method. Refs. [23,22,27,28] report successful implementation of the solution structure method with classical and trigonometric polynomials, as well as B-splines defined over uniform and non-uniform Cartesian grids. Providing exact treatment to the boundary conditions, the method can employ any suitable solution technique to compute numerical values of the degrees of freedom in the solution. And last, but not least, the solution structure method enables complete automation of the solution procedure [29,23,24].

1.1. Outline

The rest of the paper is organized as follows: Section 2 provides basic information on the solution structure method. It describes how the solution structures are constructed and used to satisfy the given boundary conditions. It also explains how the standard solution techniques have to be modified in order to accommodate solution structures. In Section 3 we describe a geometrically adaptive integration technique and its modification for handling the imprecise geometric models. The proposed engineering analysis approach is tested on a number of benchmark problems which are presented in Section 4. Section 5 summarizes advantages and weaknesses of the proposed approach and discusses the future directions. In this paper we present solutions of 2D example problems, but the proposed method can be easily generalized in 3D.

2. Solution structure method

2.1. Boundary value problem and its solution structure

The Solution Structure Method was originated by Kantorovich in 1950s. He proposed a simple technique to satisfy Dirichlet

boundary condition

$$u|_{\partial\Omega} = \varphi, \quad (1)$$

using functions that vanish on the geometric boundaries. According to Kantorovich, a function that satisfies condition (1) can be written in the following form [30]:

$$u = \omega\Phi + \varphi, \quad (2)$$

where ω is a function that takes on zero value on the boundary $\partial\Omega$ and has non-zero gradient there; function φ extends Dirichlet boundary condition (1) inside the geometric domain Ω . Since $\omega|_{\partial\Omega} = 0$, the boundary condition (1) is satisfied at all points of the geometric boundary exactly, regardless of the chosen function Φ . The main purpose of Φ is to satisfy the governing equation of the problem. In order to achieve this Φ is represented by a linear combination of the basis functions

$$\Phi = \sum_{i=1}^n C_i \chi_i. \quad (3)$$

Being substituted into (2), Φ results in a set of functions which satisfy the boundary condition (1) exactly regardless of the numerical values of the degrees of freedom $\{C_i\}_{i=1}^n$. To find an approximate solution to the boundary value problem, any suitable numerical method can be used. Usually application of a numerical method result in an algebraic system whose solution assigns numerical values to the degrees of freedom.

Kantorovich method had limited applicability mostly because it was not clear how to construct function ω that vanishes on the boundaries of complex geometric objects. With invention of R -functions by Rvachev [31,32] in 1963 Kantorovich method was reborn. Using R -functions it was possible to construct such functions for complex geometric domains represented either by Constructive Solid Geometry or Boundary Representation.

Another step forward in the development of the Solution Structure Method was understanding that approximate distance functions (fields) can be used as ω functions. Approximate distance fields can be constructed either by direct interpolation or least square fit to the samples of Euclidean distance to the geometric boundary [24,33]. For example, plots in Fig. 2 show isolines of the approximate distance fields that describe the boundary of the Michelangelo's David statue (Fig. 2(a)), CAD model of an anchor plate (Fig. 2(b)) and human femur (Fig. 2(c)).

Also, Rvachev and his students generalized Kantorovich method for other types of boundary conditions that include Neumann, Robin, Stefan–Boltzmann as well as interface boundary conditions [32,34,27,22,35,36]. Rvachev proposed a systematic way to construct *solution structures*—approximate solutions to the boundary value problems that satisfy the prescribed boundary conditions exactly. The main idea of Rvachev's approach was to represent solutions to the boundary value problems by powers of the functions that vanish on the geometric boundaries [22]

$$u(\omega) = u(0) + \sum_{k=1}^m \frac{1}{k!} u_k(0) \omega^k + \omega^{k+1} \Phi. \quad (4)$$

This representation is a straightforward generalization of a classical Taylor series, where the term $|x-x_0|$, measuring the distance to the point x_0 , is replaced by ω measuring the distance to a set of points. Similarly, the k th order derivatives of the function u in the classical Taylor series are replaced by coefficients u_k that are k th order derivatives of the function u in the direction n normal to the boundary of a geometric domain. In contrast with classical Taylor series, where the coefficients are constants, $u_k(x,y,z)$ in the expression (4) may be arbitrary functions. This also holds when ω represents *approximate* distance to the geometric boundary. Taylor series (4) provides a connection

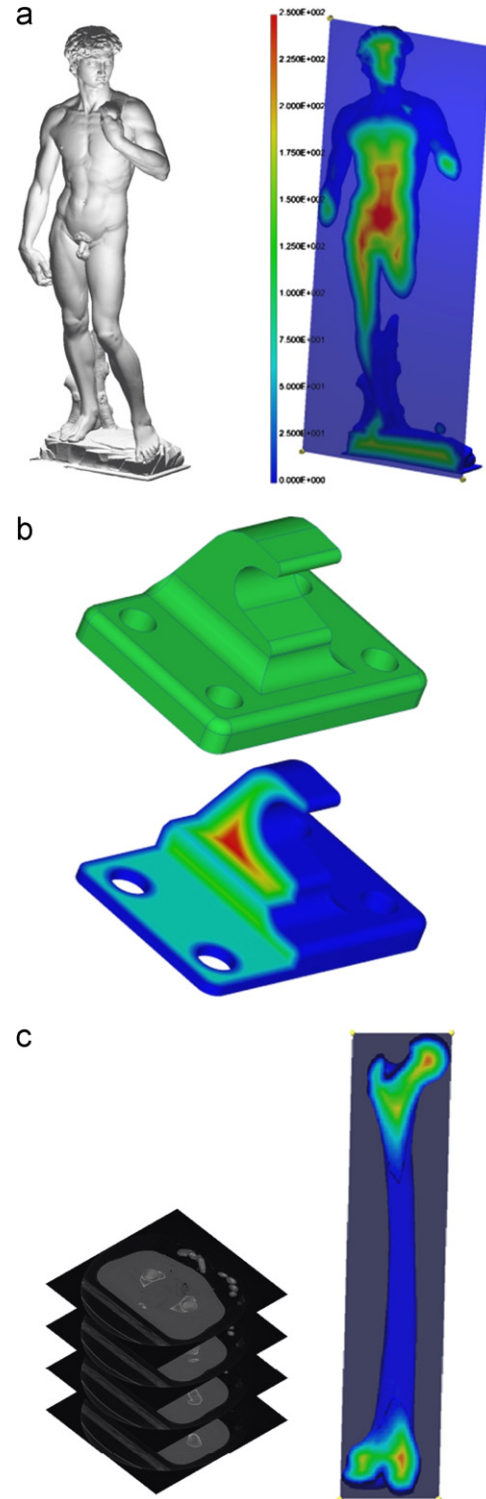


Fig. 2. Approximate distance fields can be constructed automatically from virtually all geometric representations: (a) triangulated surface of the Michelangelo's David statue and an approximate distance field to it. Blue color corresponds to the boundary points (distance is zero there). (b) CAD model and constructed approximate distance field to its surface. Distance field is shown in a cross-section. (c) Approximate distance field to the surface of a human femur constructed from segmented Computed Tomography scans. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

between the value of a physical field at any spatial point and values of the field and its normal derivatives prescribed on the boundary of a geometric domain. In the context of engineering

analysis this means that the function u given by expression (4) satisfies specified boundary conditions *exactly*.

Generalized Taylor series (4) are used to derive solution structures for different types of boundary conditions. For example, solution structure for the Neumann boundary condition

$$\left. \frac{\partial u}{\partial n} \right|_{\partial\Omega} = \varphi, \quad (5)$$

can be written as follows:

$$u = \Phi_1 - \omega D_1^\omega(\Phi_1) + \varphi\omega + \omega^2\Phi_2, \quad (6)$$

where $D_1^\omega() = \nabla\omega \cdot \nabla()$ is the first order differential operator in the direction normal to the boundary. The first two terms in expression (6) represent the unspecified value of the solution u on the domain's boundary. Subtraction of $\omega D_1^\omega(\Phi_1)$ from Φ_1 assures that the first normal derivative of the difference vanishes on $\partial\Omega$ [22]. Product of the prescribed function φ and approximate distance field ω takes care of the boundary condition (5), while remainder the term $\omega^2\Phi_2$ guarantees completeness of the solution structure (6). Undetermined functions Φ_1 and Φ_2 are usually represented by a linear combination of the basis functions (3). In the case when basis functions with a compact support, such as, for example, tensor product B-splines, are used, the last term in the solution structure, $\omega^2\Phi_2$ can be omitted. Similarly, the solution structure that satisfies Robin boundary condition

$$\left(\frac{\partial u}{\partial n} + hu \right) \Big|_{\partial\Omega} = \psi. \quad (7)$$

can be written as follows:

$$u = \Phi - \omega D_1^\omega(\Phi) + \psi\omega - h\omega\Phi. \quad (8)$$

In the context of heat transfer problems, functions h and ψ in the boundary condition (7) incorporate the ambient temperature of the environment $T_{amb}(p)$, thermal conductivity $\lambda(p)$, and convective coefficient $\alpha(p)$: $h = \alpha(p)/\lambda(p)$ and $\psi(p) = \alpha(p)T_{amb}(p)/\lambda(p)$, which are the functions of spatial coordinates.

When functions φ , h and ψ in the boundary conditions (1), (5), (7) are specified in a piecewise manner, i.e. different functions are prescribed over different pieces of the boundary, these functions can be combined into a single function that transfinately interpolates the given functions [35]. Transfinite interpolation combines approximate distance fields ω_i to the portions $\partial\Omega_i$ of the geometric boundary $\partial\Omega$ with the prescribed functions φ_i [35]

$$\varphi = \sum_{i=1}^n \varphi_i \frac{\prod_{j=1, j \neq i}^n \omega_j^{\mu_j}}{\sum_{k=1}^n \prod_{j=1, j \neq k}^n \omega_j^{\mu_j}}, \quad (9)$$

where exponents μ_j are used for interpolating the functions together with their derivatives. More information on transfinite interpolation and its properties can be found in [35]. Transfinite interpolation can be applied to the solution structures as well. For example, the solution structure for mixed boundary conditions

$$u|_{\partial\Omega_1} = \varphi; \quad \left(\frac{\partial u}{\partial n} + hu \right) \Big|_{\partial\Omega_2} = \psi \quad (10)$$

is obtained via transfinite interpolation of the individual solution structures

$$u = \frac{1}{\omega_1 + \omega_2^2} ((\omega_1\Phi_1 + \varphi)\omega_2^2 + (\Phi_2 - \omega_2 D_1^{\omega_2}(\Phi_2) + \psi\omega_2 - h\omega_2\Phi_2)\omega_1), \quad (11)$$

where ω_1 and ω_2 are approximate distance fields to the boundaries $\partial\Omega_1$ and $\partial\Omega_2$ respectively. Sometimes another solution structure that satisfies mixed boundary conditions (10) is used

$$u = \omega_1\Phi + \varphi - \omega D_1^{\omega_2}(\omega_1\Phi) - \omega D_1^{\omega_1}(\varphi) - h\omega_1\omega\Phi - h\varphi\omega + \psi\omega, \quad (12)$$

where ω is a distance field to the whole geometric boundary, and ω_1 is a distance field to the Dirichlet boundary.

2.2. Treatment of the differential equation

As we have discussed earlier, using solution structures to represent the physical fields being modeled enables exact treatment of the prescribed boundary conditions. The governing equation of a problem can be satisfied by a proper choice of the undetermined functions Φ , Φ_1 and Φ_2 in the solution structures. If these functions can be determined exactly—we have the exact solution of the problem. Unfortunately, it can be done only in a few special cases. In practical situations, however, the undetermined functions are approximated by a linear combination of the basis functions $\{\chi_i\}_{i=1}^N$ (3). In this case each solution structure defines a set of functions that satisfy the prescribed boundary conditions. Application of an appropriate solution method makes it possible to determine numerical values of the degrees of freedom in (3) such that the solution structure provides the best approximation to the governing equation of the problem. Solution structure method does not restrict the choice of the basis functions or the solution method used to compute numerical value of the degrees of freedom. It has been successfully implemented using polynomials (classical and trigonometric), tensor product B-splines as well as finite element basis functions in conjunction with Galerkin, least square and collocation methods [37,38,34,27,24,36,39]. For example, Ritz–Galerkin method is often used for solving heat conduction problems. Applied to the Laplace equation $\nabla^2 u = 0$ with mixed boundary conditions (10) combination of the solution structure and Ritz–Galerkin methods produces a system of linear algebraic equations $\mathbf{AC} = \mathbf{B}$ in which the elements of the conductivity matrix and the load vector are computed using the following expressions:

$$\begin{aligned} a_{ij} &= \int_{\Omega} \nabla \xi_i \nabla \xi_j \, d\Omega + \int_{\partial\Omega_2} h \xi_i \xi_j \, dS; \\ b_i &= - \int_{\Omega} \nabla \xi_i \nabla u_1 \, d\Omega + \int_{\partial\Omega_2} \xi_i (\psi - hu_1) \, dS, \end{aligned} \quad (13)$$

where $u_1 = \varphi - \omega D_1^{\omega_2}(\varphi) - h\varphi\omega + \psi\omega$ is a portion of the solution structure which satisfies non-homogeneous boundary conditions. The solution structure operator transforms the chosen basis functions $\{\chi_i\}_{i=1}^N$ into a new system of basis functions $\{\xi_i\}_{i=1}^N$ that satisfy homogeneous boundary conditions of the same type

$$\xi_i = \omega_1 \chi_i - \omega D_1^{\omega_2}(\omega_1 \chi_i) - h\omega_1 \omega \chi_i.$$

Application of the solution structure method to structural analysis problems is often formulated as balance between the work performed by the external forces and the potential energy of the elastic deformation

$$\int_{\Omega} \boldsymbol{\epsilon}^T \boldsymbol{\sigma} \, d\Omega = \int_{\Omega} \mathbf{u}^T \mathbf{F} \, d\Omega + \int_{\Gamma_t} \mathbf{u}^T \mathbf{T} \, d\Gamma_t, \quad (14)$$

where $\mathbf{u} = (u_x, u_y, u_z)^T$ is a displacement vector, strain $\boldsymbol{\epsilon}$ and stress $\boldsymbol{\sigma}$ are second order tensor quantities represented by six-dimensional vectors according to the Voigt notation. Boundary conditions are formulated as the prescribed displacements specified over boundary Γ_u and the traction load specified on the boundary Γ_t . In addition to the boundary forces the geometric model can be subject to the body (gravity) force $\mathbf{F}$.

To satisfy kinematic (essential) boundary condition displacement $\mathbf{u}$ can be represented by the following solution structure:

$$\mathbf{u} = \sum_{i=1}^n \mathbf{C}_i(\omega^x, \omega^y, \omega^z)^T \chi_i + \mathbf{u}^*, \quad (15)$$

where $\omega^x, \omega^y, \omega^z$ are functions that vanish on the boundaries on which essential boundary conditions are prescribed; and vector

valued function $\mathbf{u}^* = (u_x^*, u_y^*, u_z^*)^T$ interpolates all non-zero displacements in x , y and z coordinate directions respectively. Designating $(\omega^x, \omega^y, \omega^z)^T \chi_i$ as the new basis functions $\boldsymbol{\eta}_i$ that satisfy homogeneous Dirichlet (essential) boundary conditions, the system of the linear algebraic equations, from which numerical values of the degrees of freedom $\mathbf{C}_i$ are determined, can be written as follows [24]:

$$\begin{aligned}
 - \sum_{i=1}^n \mathbf{C}_i \underbrace{\int_{\Omega} \mathbf{B}^T[\boldsymbol{\eta}_i] \mathbf{D} \mathbf{B}[\boldsymbol{\eta}_j] d\Omega}_{\text{stiffness coefficient}} &= \underbrace{- \int_{\Omega} \boldsymbol{\eta}_j \mathbf{F} d\Omega}_{\text{load due to body force}} \\
 + \underbrace{\int_{\Omega} \mathbf{B}^T[\boldsymbol{\eta}_j] \mathbf{D} \mathbf{B}[\mathbf{u}^*] d\Omega}_{\text{load due to applied displacements}} &- \underbrace{\int_{\Gamma_t} \boldsymbol{\eta}_j \mathbf{T} d\Gamma_t}_{\text{load due to applied loads}}
 \end{aligned} \quad (16)$$

where $\mathbf{B}$ denotes the matrix of derivatives, also known as the strain–displacement matrix, so that $\boldsymbol{\epsilon} = \mathbf{B}[\mathbf{u}]$, and $\mathbf{D}$ represents the stress–strain matrix so that $\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\epsilon}$. Weak formulations of boundary value problems (13), (16) require numerical integration over the given geometric domain which is performed by using the integration lattice rules that place integration points inside the geometric model.

Besides weak formulations, solution structure method can be used in conjunction with the collocation method [39]. In this case solution structure is used to satisfy the prescribed boundary conditions while the governing equation is enforced at the collocation points allocated inside the geometric domain. Our numerical experiments suggest that using solution structures to enforce the prescribed boundary conditions improves convergence of the approximate solutions and may result in a better conditioned algebraic system [40].

2.3. Examples of application of SSM to engineering problems

Let us demonstrate an application of the Solution Structure Method to heat conduction and plane stress analysis problems.

Heat transfer. First, let us consider heat conduction in a 2D heat sink which is shown in Fig. 3(a). A temperature of 100 °C is specified on the boundary $\partial\Omega_1$. On the rest of boundaries convective heat exchange is given

$$T|_{\partial\Omega_1} = 100; \quad \left(\frac{\partial T}{\partial n} + \frac{\alpha}{\lambda} T \right) \Big|_{\partial\Omega_2} = \frac{\alpha T_{amb}}{\lambda}. \quad (17)$$

The specified values of the convection coefficient, α and ambient temperature, T_{amb} are 10 W/m² K and 20 °C respectively. The heat conductivity, λ , of the material (Aluminum) is 200 W/m K. The SSM solution procedure starts with analyzing the type of the boundary conditions and construction of the solution structure that enforces them. In our case we will represent temperature, T , using solution structure (11) that satisfies the mixed boundary conditions (17). The solution structure (11) contains approximate distance fields ω_1 and ω_2 that vanish on the boundaries where Dirichlet and Robin boundary conditions are prescribed. These functions are constructed using the original geometric representation. In our case ω_1 and ω_2 are constructed from Boundary Representation of the geometric domain in Fig. 3(a). Plots in Fig. 3(b) and (c) present isolines of ω_1 and ω_2 respectively. Blue color in these plots illustrates the position of the zero set of the approximate distance fields. Once approximate distance fields and solution structure are constructed, application of Galerkin method results in an algebraic system $AC=B$ in which elements of the matrix A and vector B are computed using expression (13). Numerical solution of the algebraic problem delivers numerical values of the degrees of freedom in the solution structure. Finally, the solution, T , can be computed and visualized. Fig. 3(d) illustrates distribution of the computed temperature.

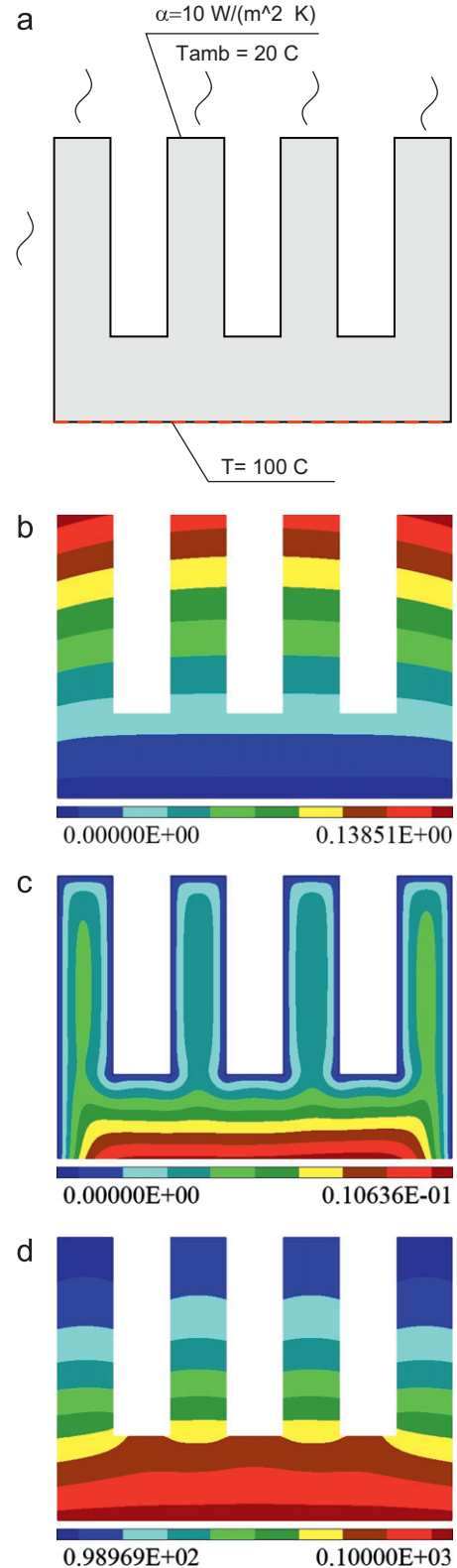


Fig. 3. Heat transfer in a 2D heat sink: (a) Temperature of 100 °C is specified at the bottom edge of the geometric domain, and convective heat exchange is given on the rest of the boundaries. Approximate distance fields to (b) Dirichlet and (c) convective boundaries. (d) Temperature distribution predicted by the solution structure method. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

Stress analysis. The next example illustrates application of the solution structure method to plane stress analysis problems. Geometry of a cantilever beam, applied load and fixations are

shown in Fig. 4. Geometric boundary shown in red color is fixed in both coordinate directions while the green boundary is fixed in the horizontal direction only. To satisfy the kinematic boundary conditions the components of the displacement vector are represented by the following solution structures:

$$u_1 = \omega_1 \sum_{i=1}^N C_i^1 \chi_i; \quad u_2 = \omega_2 \sum_{i=1}^N C_i^2 \chi_i. \quad (18)$$

As in the previous example, approximate distance fields ω_1 and ω_2 are constructed from Boundary Representation of the geometric domain shown in Fig. 4. Their isolines are shown in Fig. 5(a) and (b). Blue color in these plots correspond to the boundaries whose motion is restricted in horizontal and vertical directions. The basis functions χ_i are chosen to be tensor product B-splines defined over 60×30 uniform Cartesian grid. Application of the solution method results in numerical values of the degrees of freedom C_i^1 and C_i^2 . Substituting them back into solution structures (18) we can evaluate and visualize the components of the displacement vector as well as the distribution of the von Mises stress (see plots in Fig. 6).

2.4. Observations

As we have seen the solution method, which enforces the governing equation of the problem, requires sampling integration or collocation points *inside* the geometric model. It can be done relatively easily if the geometric model constitutes a valid solid model. In the presence of the geometric imperfections (holes, dangling boundaries, self-intersecting boundaries, etc.) notions of “inside” and “outside” are not defined. As a result this precludes application of the solution method that enforces the governing equation.

In the next section, we will describe a geometrically adaptive integration technique that can be used to sample integration/collocation points inside geometric domains with geometric

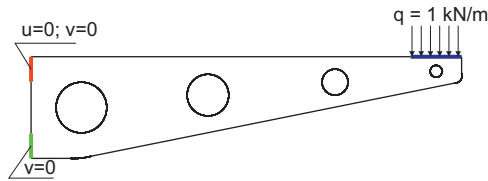


Fig. 4. Loaded 2D cantilever beam. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

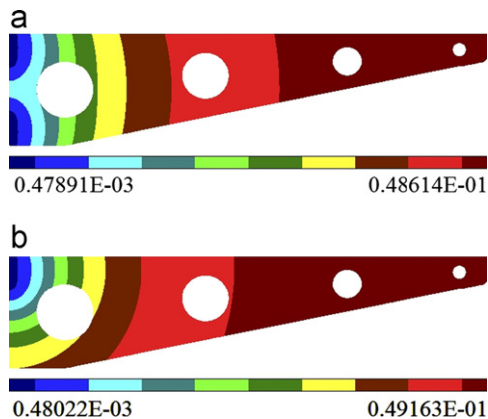


Fig. 5. Distance fields to the fixed boundaries: (a) in a horizontal direction and (b) in vertical direction. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

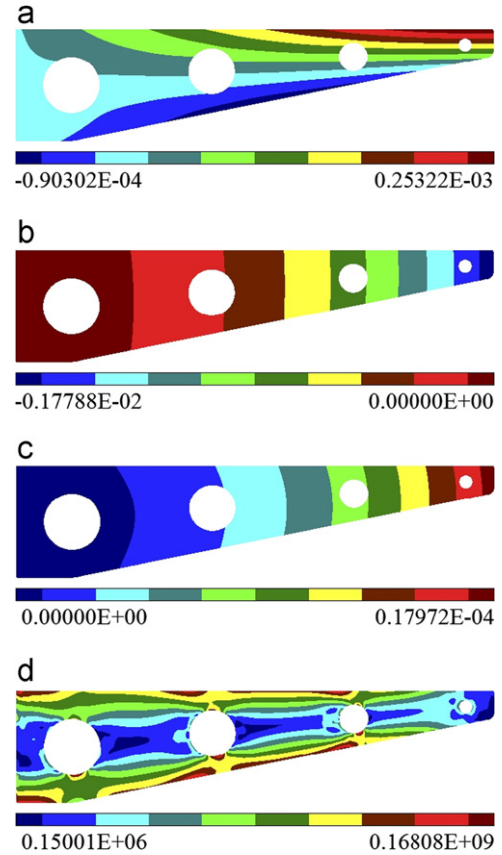


Fig. 6. Computational results provided by the solution structure method: (a) horizontal component of the displacement vector; (b) vertical component of the displacement vector; (c) magnitude of the displacement; (d) von Mises stress.

imperfections without performing computationally expensive geometry healing.

3. Geometrically adaptive integration over domains with geometric imperfections

3.1. Geometrically adaptive integration

In [23,25] we described a geometrically adaptive volumetric integration technique which can be used in conjunction with the solution structure method to assemble the algebraic system. The integration process starts with an allocation of a Cartesian grid over the geometric model. The grid is constructed in a such way that the its cells are aligned with the supports of the basis functions (Fig. 7(a)). Once cells are allocated, they are classified with respect to their position relative to the geometric model. This classification splits the cells into three groups: external, internal, and geometry cells. It can be performed by analyzing the positions of the cells vertices with respect to the geometric model. For example, all vertices of the external cells have to be located outside of the geometric model. Similarly, all vertices of an internal cell must be situated inside the geometric model. Geometry cells have some of their vertices inside the geometric model and some outside. A grid cell can be also classified as a geometry cell if it encloses the boundaries of a geometric domain while having all its vertices inside or outside of the geometric domain.

Once the classification phase is over, the integration points are allocated in internal and geometry cells, and numerical integration is performed. During the integration phase external cells are

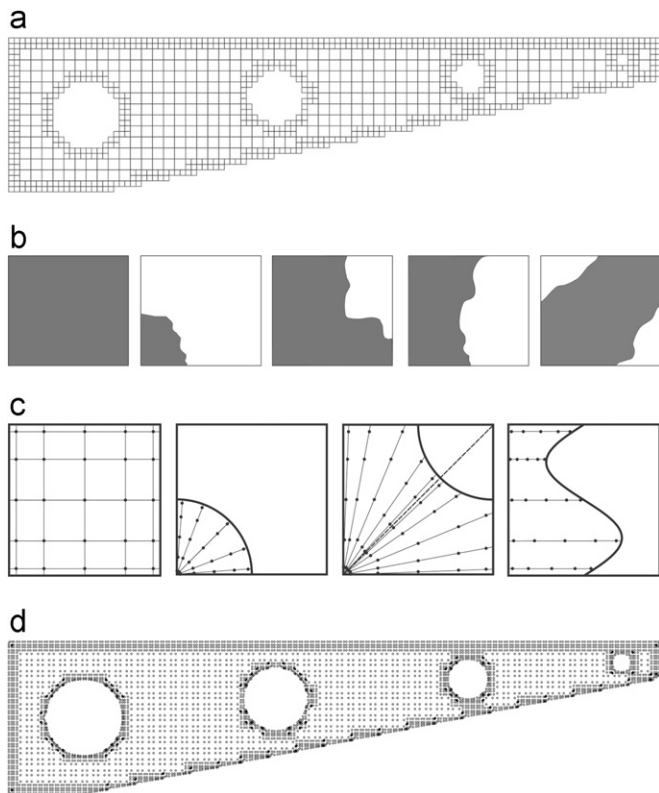


Fig. 7. (a) Hierarchical space decomposition is imposed over the geometric domain in Fig. 4. (b) Basic types of the 2D intersection cases (up to symmetry and rotation transformation). (c) Allocation of the integration points in the integration cells. (d) Positions of the integration points in the geometric domain shown in Fig. 4.

simply ignored. Since internal cells are rectangles in 2D and cuboids 3D, the integration points in these cells are placed in accordance with lattice integration rules [41]. Allocation of the integration points in geometry cells, due to their arbitrary geometric shape, requires special attention. As we demonstrated in [25], a proper space parametrization in geometry cells can substantially reduce the integration error. The choice of the space parametrization depends on how the geometric boundary intersects the integration cell [23,25]. In our implementation of the geometrically adaptive integration we use a cell classification similar to the one employed by the marching cubes technique [13]. Basic types of the 2D intersection cases (up to symmetry and rotation transformation) are presented in Fig. 7(b). Depending on the type of the intersection case different space parameterizations are employed. For example, a polar coordinate system is used if the integration cell has one or three vertices inside the geometric domain, as it is shown in Fig. 7(c). In the other cases Cartesian coordinates are used to allocate the integration points. To integrate in 3D space spherical, cylindrical and Cartesian coordinate systems are employed. Some intersection cases are ambiguous (like the right-most case in Fig. 7(b))—the positions of internal and external vertices allow multiple intersection configurations. In such cases the ambiguity is resolved by applying hierarchical subdivisions to the cell. The cells are also hierarchically subdivided if they contain small geometric features [25]. Fig. 7(d) illustrates allocation of the integration points inside geometric domain shown in Fig. 4.

This geometrically adaptive integration assumes that the geometric model provides the ability to compute a Point Membership Classification (PMC)—a function that returns a relative position of a given point in the space with respect to the geometric model. Most geometric engines support PMC

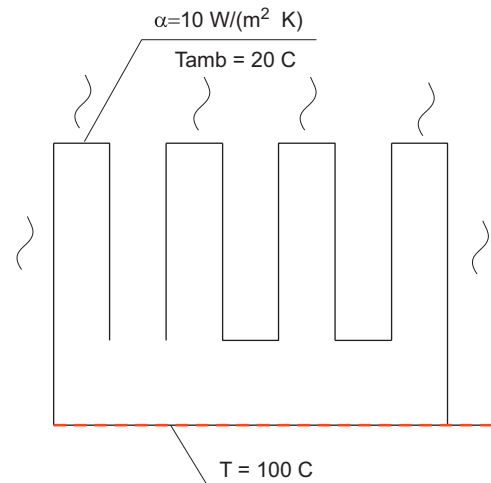


Fig. 8. A 2D steady-state heat transfer problem with imperfect geometry.

computations for valid solid models. In the presence of geometric imperfections direct computation of the PMC becomes impossible and the geometric model requires healing.

Now we will explain how our geometrically adaptive integration should be modified in order to perform volumetric integration over geometrically inaccurate models without application of computationally expensive healing techniques.

3.2. Integration over domains with geometric imperfections

In order to place integration nodes in geometrically imprecise models, the PMC values have to be computed at the nodes of a hierarchical space decomposition. The presence of gaps and self-intersections of the geometric boundary cause PMC computation to fail. Since traditional methods of repairing geometric models can be computationally expensive, we propose a method of handling geometric imperfections which does not alter the original geometry. In this work, a hybrid geometric model is employed, composed of a hierarchical space decomposition, Boundary representation (B-rep) and distance fields. The idea of the proposed approach is quite simple: in order to enable PMC computations the geometric domain needs to be closed in one of these available representations. In our implementation we use hierarchical space decomposition to resolve all geometric inaccuracies, while the original unrepaired model is used later to allocate the integration points.

The integration process over imprecise geometric models starts with the construction of a Cartesian grid inside the bounding box that encloses the geometric model (Fig. 9(a)). Once the grid is constructed, the geometric boundaries are rasterized (pixelized in 2D and voxelized in 3D) over the grid cells using scan-conversion algorithms [42]. In our data structure the cells which contain or intersect with the boundaries of the geometric model are marked as the *geometry cells*. In Fig. 9, geometry cells are shown in gray and red colors. To increase the accuracy of the rasterized representation, the geometry cells can be hierarchically subdivided into subcells. If the original geometric model contains geometric defects or imprecisions, they are also inherited by the rasterized representation. To remove geometric defects and enable PMC computations at the nodes of a hierarchical space decomposition the geometry cells are further classified into *regular* and *irregular* cells. Edges of regular cells have an even number of intersections with the geometric boundaries. In addition, each edge of a regular cell may not intersect the boundary more than once. These conditions correspond to the continuous

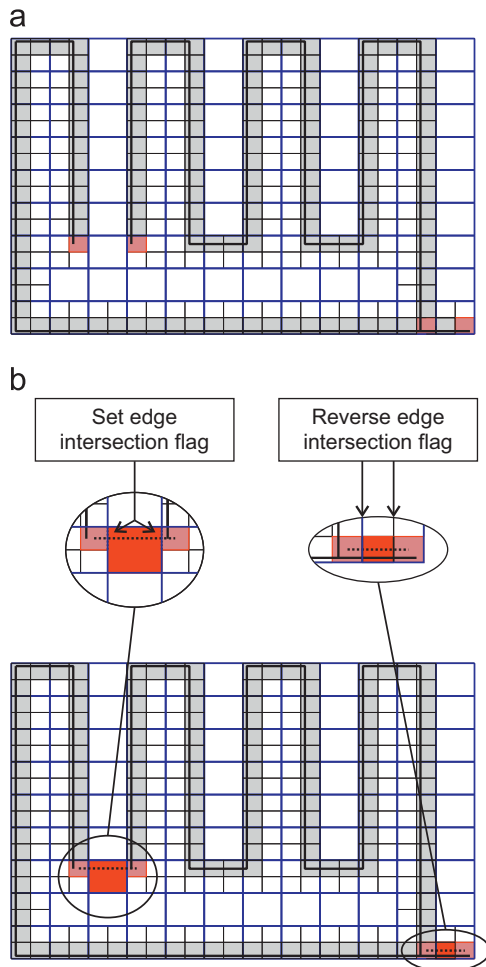


Fig. 9. (a) Heatsink with initial Cartesian grid (blue), one application of geometry cell subdivision and marked regular (gray) and irregular geometry cells (light red). (b) Closure of gaps and non-manifold boundary removal using hierarchical space decomposition: closing lines (dashed lines) are converted into the closing cells (dark red). Non-manifold boundaries are removed by reversing the value of the intersection flag for cell edges. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

boundary—the boundary comes in to the cell and goes out. Regular cells appear in gray color in Fig. 9. In the cases when cell's edges have multiple intersections with the geometric boundary such cells are hierarchically subdivided until this artifact is removed. Irregular geometry cells (shown in light red color in Fig. 9) mark the locations of the geometric defects, such as, for example, gaps and non-manifold boundaries and have odd number of intersections with the boundary of the geometric model.

After hierarchical space decomposition is built, we cycle through the list of irregular geometry cells and connect each closest-pair with a line, which will be referred to as the *closing line*. For brevity, we connect the cells center-to-center, though there is no restriction on the end point locations. Once a closing line has been generated, it is rasterized using scan-conversion algorithms on the existing hierarchical space decomposition. All grid cells that intersect the closing line will be referred to as *closing cells* and the edge intersection flags of these cells are set appropriately. When cell's edge intersects the boundary of the original geometric domain, the value of the intersection flag of that edge reflects the number of the intersection points. When the closing line crosses the cell's edge, it changes the value of the intersection flag: the flag is incremented if the edge has

previously had no intersections, and decremented otherwise. This enables closing of gaps, and removal of non-manifold boundary pieces (see Fig. 9(b)) and resolution of the overlapping boundaries (Fig. 10). This 2-step process of generate-then-intersect is applied to all pairs of irregular geometry cells. It is important to point out two aspects of this method: first, the original geometric definition remains *unchanged* and second, each closing line is generated locally in the hierarchical space decomposition and discarded immediately after placement of integration nodes.

Once the geometric model is closed in the hierarchical space decomposition representation, PMC computation can proceed. It starts from one of the vertices of the bounding box at which the PMC value is specified by the user. For definiteness, let us start from the left bottom corner of the bounding box. Also let us assume that this vertex lies outside of the geometric model. Now we will take a look at a cell of the hierarchical space decomposition that contains that vertex (Fig. 11). Starting from the point at which the PMC value is known we visit other vertices of the cell, taking into account whether or not the cell edges intersect the boundary or a closing line. For example, the intersection flag for the edge AB (Fig. 11) is not set. This means that the point B has the same PMC value as the point A. When the intersection flag is set, like for the edge BC, the PMC values at the end points of the edge have the opposite values. This means that PMC value at the point C should be set to “inside”. Similarly, because the edge AD does not intersect neither geometric boundary nor a closing line, the PMC value the point D coincides with the PMC value at the point A. Once PMC values at all vertices of a cell are determined, they are propagated to the neighboring cells in a similar manner. Fig. 12(a) illustrates the final distribution of the PMC values at the nodes of the hierarchical space decomposition.

In the case of two or more non-manifold boundaries attached to the model's boundary at the same point (Figs. 13(a) and 15(a)), the proposed method creates an “attachment” that can be observed in Fig. 13(a). Creation of such attachment can be prevented if two additional irregular geometry cells are placed in the cell which has four edge-boundary intersections as illustrated in Fig. 13(b). The closing lines connect irregular cells with one edge-boundary intersection with the irregular cells with four edge-boundary intersections. The closing lines passing through the cells which contain the non-manifold boundary pieces reverse intersection flags on the cells edges. As a result the non-manifold



Fig. 10. The closing line decrements the values of the edge intersection flag. This helps resolving the overlapping boundaries.

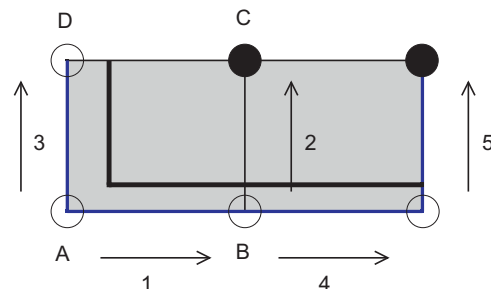


Fig. 11. Propagation of the PMC values.

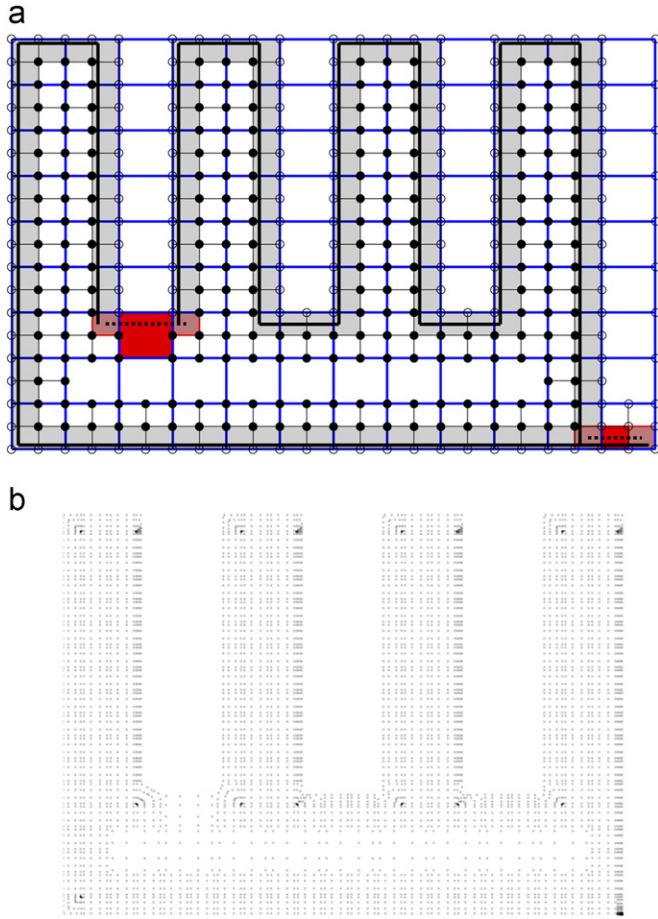


Fig. 12. (a) PMC values computed for the geometric model in Fig. 8. Filled circles designate internal points, while non-filled ones correspond to the points outside the geometric model. (b) Integration points are placed using both original geometric model and closing lines.

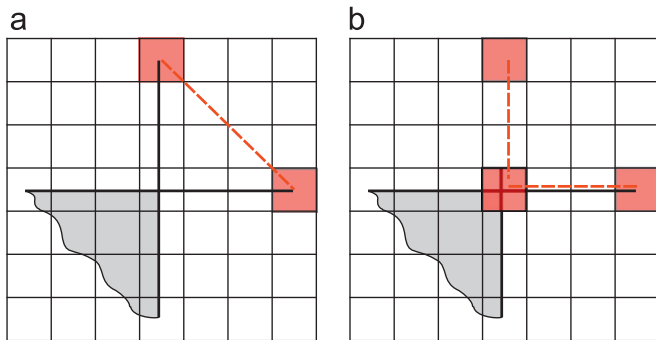


Fig. 13. Resolution of non-manifold boundaries: (a) placement of two irregular cells resolves non-manifold boundaries but creates an “attachment” to the geometric model. (b) By placing two additional irregular cells in the cell which has four edge-boundary intersections non-manifold boundary pieces can be removed.

boundary pieces are removed and an attachment is not created (Fig. 13(b)).

After the PMC values of all grid nodes have been determined, allocation of integration nodes can begin. For internal cells and regular geometry cells, integration nodes are allocated in the same manner as with perfect geometric models. Irregular geometry cells and closing cells can be treated as external cells (no integration nodes), internal cells (filled with integration nodes) or as regular geometry cells (partially filled with nodes).

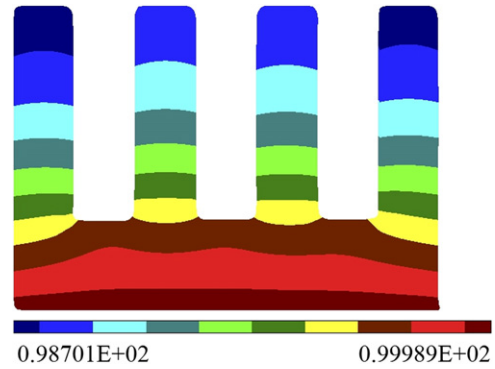


Fig. 14. Distribution of the temperature field in imprecise geometric model shown in Fig. 8.

In this work, we use the third option—both the geometry and the closing line(s) are used to allocate integration nodes. In the closing cells, integration nodes are allocated as in regular geometry cells where the “geometry” is the closing line. In the irregular geometry cells, integration nodes are allocated using a line whose end-points are the points where the geometry and closing line intersect the cell. It is important to note that though other end-point combinations are possible, our combination is easily carried into 3D problems. Fig. 12(b) illustrates the final placement of the integration nodes. The plot of the resulting temperature distribution is presented in Fig. 14.

4. Numerical experiments

The main purpose of the numerical experiments discussed in this section is to investigate the behavior of the proposed analysis technique in the presence of several different types of imperfections as well as the sensitivity of the modeling results to the size of the geometric imperfections. As before we will limit our discussion to consideration of heat transfer and structural analysis problems.

4.1. Heat conduction in imperfect geometric model of a heat sink

Let us start with considering a steady heat transfer problem in the geometric domain shown in Fig. 15(a). In this example we will use the same boundary conditions and material properties as in the heat transfer example in Section 2.3. This time, however, the geometric model has four gaps of different size, two overlapping boundary pieces, and three non-manifold boundary pieces. Plots in Figs. 16 and 17 illustrate the temperature distributions obtained using the proposed approach. In this numerical experiment coarse 35×35 (Fig. 16) and fine 70×70 (Fig. 17) Cartesian grids of biquadratic B-splines have been used. Temperature distributions shown in Figs. 16(a) and 17(a) were obtained without hierarchical subdivision of the geometry cells.

To model the temperature distributions shown in Figs. 16(b) and 17(b) the geometry cells were twice subdivided. Fig. 15(b) illustrates resolution of the geometric imperfections on a coarse 35×35 grid. Non-manifold boundary in bottom-right corner was removed and gaps were closed according to the algorithm we described in Section 3.2. The closing lines in Fig. 15(b) are shown with red dashed line. The algorithm removed two non-manifold boundaries in upper-left corner of the geometric model and resolved two overlapping boundaries. Fig. 16(a) and (b) also illustrate allocation of the integration points near geometric imperfections.

As we can see, the original geometric boundaries as well as the closing lines participate in positioning of the integration points.

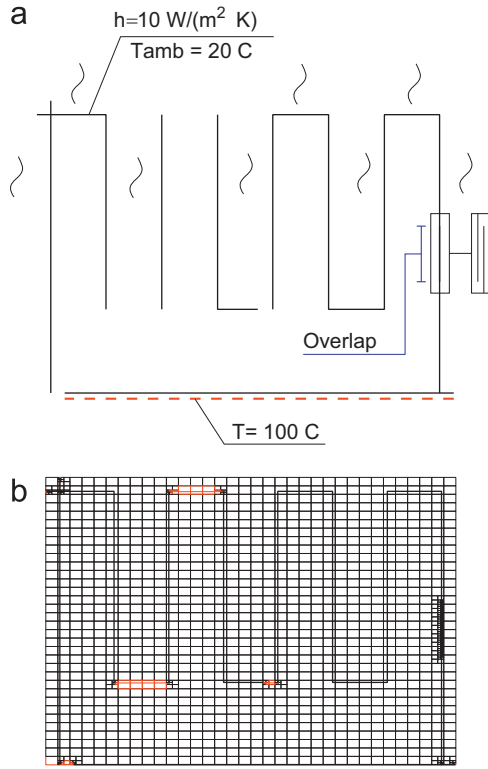


Fig. 15. Heat transfer in geometrically imperfect model of a heat sink: (a) geometric model and boundary conditions; and (b) resolution of geometric imperfections on a hierarchical space decomposition.

Two levels of hierarchical subdivisions of the geometry cells (Fig. 16(b)) result in denser cloud of the integration points near original geometric boundaries. Denser grids of basis functions and/or subdivisions of the geometry cells result in converging numerical solutions. Comparison of temperature distribution in Fig. 3(d) with those in Figs. 16 and 17 reveals noticeable difference between solutions obtained for perfect and imperfect geometric models. This can be explained by the fact that we assumed heat insulation on the closing boundaries. These boundary conditions are natural for Galerkin method and do not require any additional treatment. As we will demonstrate in our next example, as the size of geometric imperfections gets smaller the modeling results converge to the ones obtained in the perfect geometric models.

4.2. 2D stress analysis in geometrically imperfect cantilever beam

In this numerical experiment we will investigate behavior of the proposed analysis technique when it is applied to stress analysis problems in imperfect multiple-connected geometric domain with curvilinear boundaries. We will also explore how the numerical solution depend on the size of the geometric imperfections which include disconnected and non-manifold boundaries.

Fig. 18 illustrates an imperfect geometric domain with loads and fixations applied to its boundaries. In this numerical experiment we used two geometric models with different size of the geometric imperfections. The first model exhibits large gaps and dangling boundaries. Distributions of the magnitude of the displacement vector and von Mises stress are presented in Figs. 19 and 20, respectively. To perform computations, coarse 35×15 (Figs. 19(a, b) and 20(a, b)) and fine 75×25 (Figs. 19(c) and 20(c)) uniform Cartesian grids of biquadratic B-splines have been used. Analysis of the plots in Fig. 19 reveals that convergence of the displacements is achieved with an increase of the grid density and level of

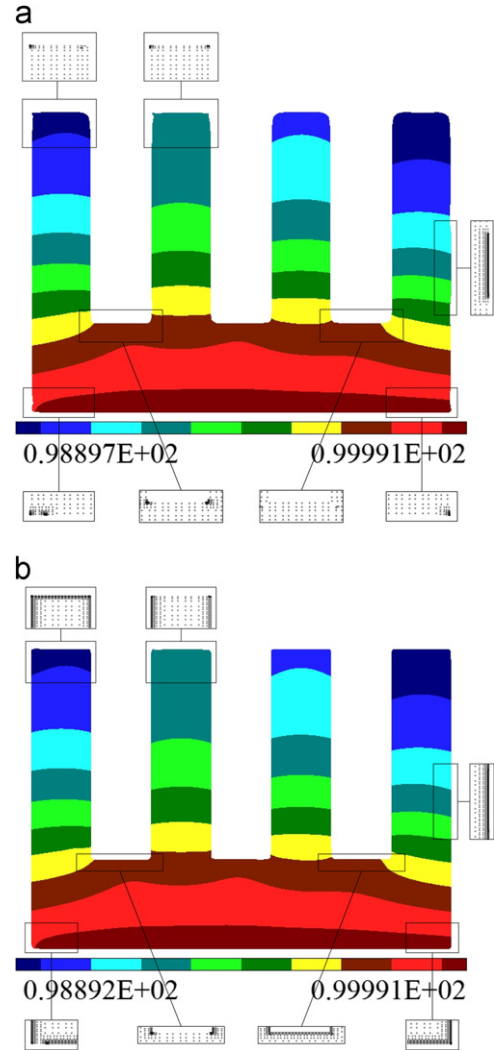


Fig. 16. Temperature distributions in imperfect geometric model shown in Fig. 15(a) obtained using biquadratic B-splines defined over uniform 35×35 (coarse) Cartesian grid: (a) without subdividing the geometry cells; (b) with two hierarchical subdivisions of the geometry cells.

hierarchical subdivision of the geometry cells. We can also see that the numerical solution with larger approximation space (Figs. 19(c) and 20(c)) provide better approximation of the stresses.

Comparing plots in Figs. 19, 20, 21 and 6 we observe that as the size of the geometric imperfections decreases the displacements in imperfect geometric domains converge to those computed in a geometrically perfect domain. Comparisons of the stress distributions illustrates that the stresses are more sensitive to the presence of the geometric imperfections since geometric defects play role of the stress concentrators. However, the presence of the geometric defects do not have a global effect on stress distributions. Stresses demonstrate elevated values locally, near the locations of the geometric imperfections.

5. Conclusions and future directions

The proposed numerical technique enables engineering analysis in imprecise geometric models without applying computationally expensive geometry or mesh repair procedures. The key feature of our approach is a modified geometrically adaptive numerical integration procedure which is based on the ability to compute PMC values at the nodes of hierarchical space decomposition even in

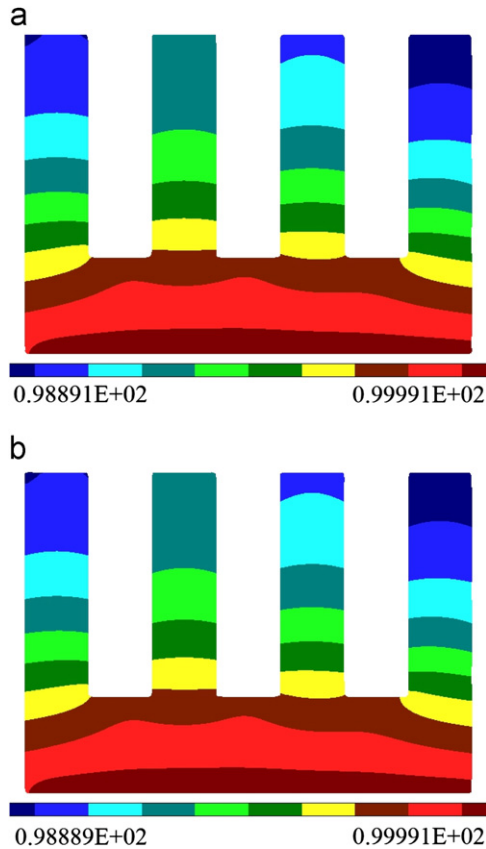


Fig. 17. Temperature distributions in imperfect geometric model shown in Fig. 15(a) obtained using biquadratic B-splines defined over uniform 70×70 (fine) Cartesian grid: (a) without subdividing the geometry cells; (b) with two hierarchical subdivisions of the geometry cells.

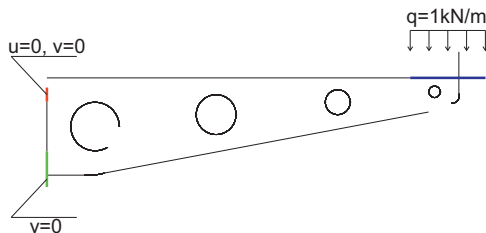


Fig. 18. 2D cantilever beam with geometric defects.

the presence of geometrical imperfections and inaccuracies. The developed algorithms use the original geometric representation wherever it is possible (in regular geometry cells). In irregular geometry cells integration points are placed based on the location of the original and closing boundaries. Distortions caused by the mismatched original and closing boundaries can be controlled by a proper selection of the subdivision level of the irregular geometry cells. The proposed numerical approach offers completely automated solution procedure for small geometric imperfections. For larger geometric inaccuracies the algorithm will still produce a solution, however, interpretation of the solution in some cases may require user intervention. The implemented algorithms automatically detect geometric imperfections and close the geometric model for the volume integration. The only information which the user has to provide is the size of the initial Cartesian grid, which is also used to construct a grid of the B-spline basis functions; and subdivision levels for the geometry cells. As an option, a different subdivision levels of the regular and irregular geometry cells can be specified. Combined with the solution structure method, our numerical

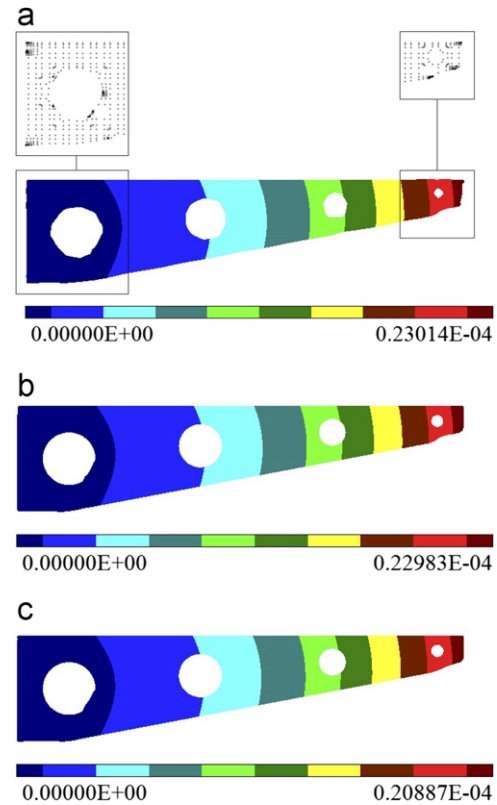


Fig. 19. Results of structural analysis of a 2D cantilever beam with large geometric defects: magnitude of the displacement vector obtained using (a) a coarse 35×15 uniform Cartesian grid of biquadratic B-splines with no subdivision of the geometry cells. (b) A coarse 35×15 uniform Cartesian grid of biquadratic B-splines with two hierarchical subdivision of the geometry cells. (c) A fine 75×25 uniform Cartesian grid of biquadratic B-splines with two hierarchical subdivision of the geometry cells.

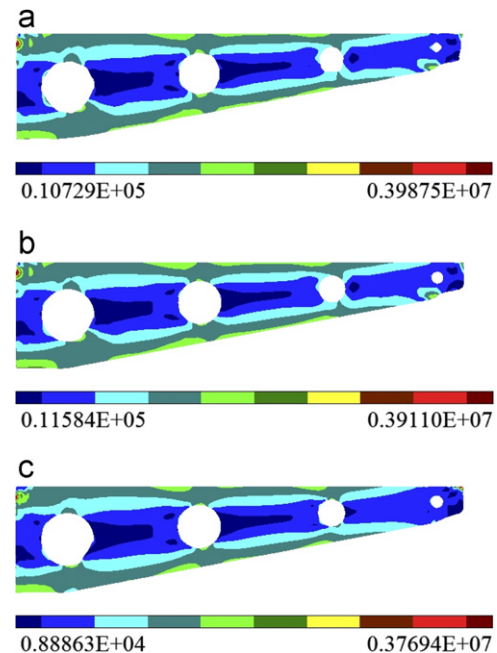


Fig. 20. Results of structural analysis of a 2D cantilever beam with large geometric defects: distribution of the von Mises stress obtained using (a) a coarse 35×15 uniform Cartesian grid of biquadratic B-splines with no subdivision of the geometry cells. (b) A coarse 35×15 uniform Cartesian grid of biquadratic B-splines with two hierarchical subdivision of the geometry cells. (c) A fine 75×25 uniform Cartesian grid of biquadratic B-splines with two hierarchical subdivision of the geometry cells.

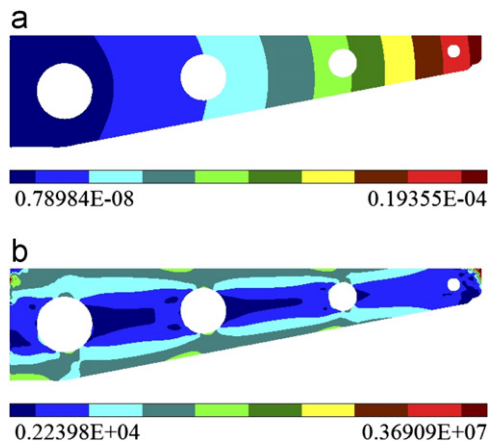


Fig. 21. Results of structural analysis of a 2D cantilever beam with small geometric defects: (a) magnitude of the displacement vector and (b) distribution of the von Mises stress obtained using a fine 75×25 uniform Cartesian grid of biquadratic B-splines with two hierarchical subdivision of the geometry cells.

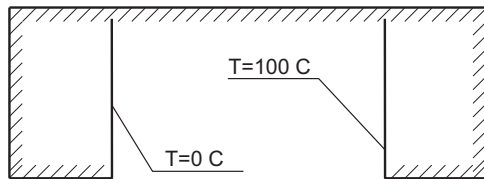


Fig. 22. Heat transfer in geometrically imperfect model.

technique makes it possible to use spatial meshes that do not necessarily conform to the shape of geometric models while enforcing the prescribed boundary conditions using solution structures as we discussed in Section 2. The meshfree nature of the method and its ability to handle geometric imperfections result in a much higher geometric flexibility in comparison with the traditional analysis tools.

Numerical examples presented in Section 4 have demonstrated high reliability in handling imperfections in the geometric models. The proposed approach can handle the majority of the real-world imperfections such as gaps, self intersecting boundaries as well as being able to remove non-manifold boundary pieces. Our volume integration algorithms were designed with an assumption that the geometric imperfections are small in comparison with the size of the geometric domain. However, the conducted numerical experiments confirmed that even large geometric imperfections can be handled by our integration algorithms.

In conclusion, we would like to draw attention to a couple of special cases which require close consideration. First, let us take a look at the geometric model in Fig. 22. This non-closed geometric model results in two irregular cells which are very close to the geometric boundary. The way how this geometric imperfection is handled depends on the size of the initial Cartesian grid and specified level of geometry cells subdivision. If the integration cells are large enough to contain both the boundary of the original geometric domain and the closing line, the latter causes elimination of such cells from the numerical integration. Fig. 23(a) illustrates the location of the integration cells, original geometry and the closing line. The simulated temperature field is shown in Fig. 23(b). It demonstrates that the geometric domain was disconnected in two rectangular parts. However, if a finer initial grid is used or if the depth of the geometry cells subdivisions is chosen such that the original geometric boundary and the closing line do not appear in the same cells, the integration algorithm closes the gap in the geometry as expected (Fig. 23(c)), and the resulting temperature field is presented in Fig. 23(d).

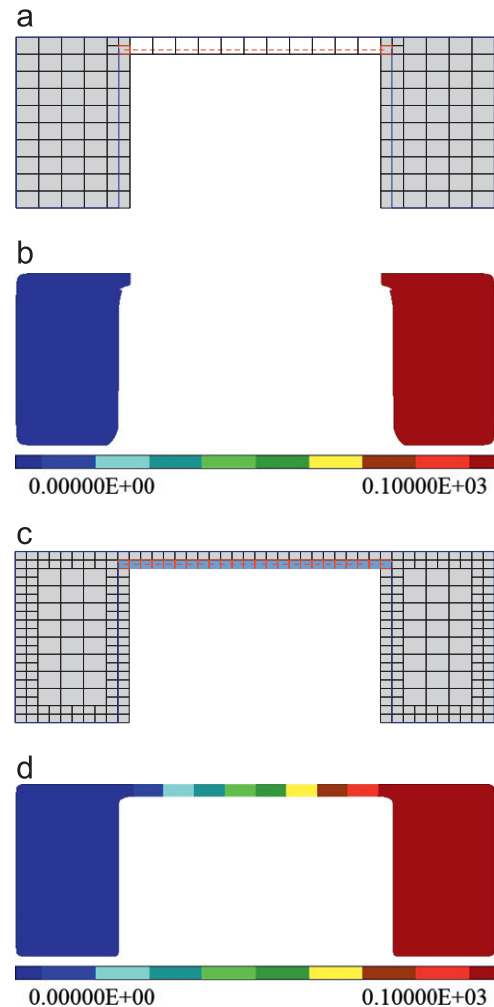


Fig. 23. Results of computer simulation of the temperature field in geometric model shown in Fig. 22: (a) coarse initial integration grid results in relatively large cells that contain the valid geometric boundary as well as the closing line. The latter removes the corresponding integration cells and splits the geometric domain into two parts. The integration cells are shown here in gray color. (b) Temperature distribution obtained using the integration cells in (a). (c) Hierarchical subdivision of the geometry cells makes it possible to separate the geometric boundary and the closing line. In this case the closing line closes the geometric boundary. The integration cells are shown here in gray color. (d) Temperature distribution obtained using the integration cells in (c).

Fig. 24(a) illustrates another special case when the closing line intersects with the boundary of the original geometric model. This case contradicts our initial assumption about small size of the geometric imperfections, and, if processed, may lead to unexpected results, as shown in Fig. 24(b). Cases such as this one, however, can be easily detected by testing whether the closing boundaries intersect with the boundary of the original geometric model. If the intersection is detected, a warning message is generated. Then the user has either to accept the modeling results (if they have a sense) or modify the geometric model in a such way that the closing boundaries do not intersect with the original geometric boundaries as it is shown in Fig. 24(c). Fig. 24(d) presents distribution of the corresponding temperature field.

When a simulation of the physical fields in imprecise geometric domains is performed, the validity of the computational results must be ensured. Our preliminary results suggest that in most cases the computation error introduced by the imperfections gets smaller when the size of the geometric imperfections decreases. In analyzing the result of our numerical experiments we also noticed that the difference between solutions obtained in a valid and imprecise

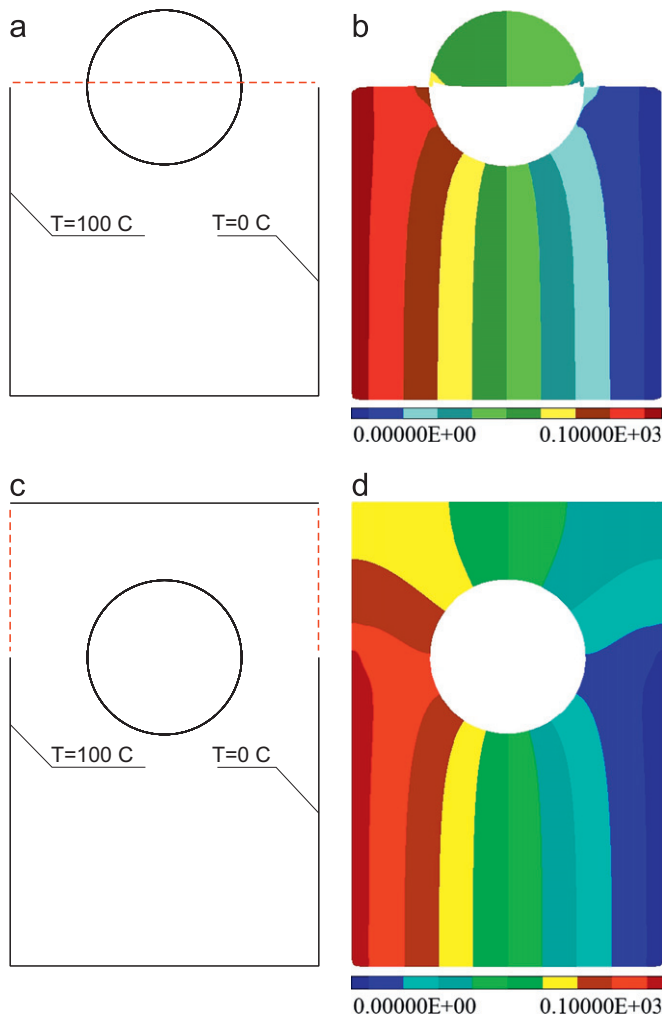


Fig. 24. (a) Closing line (shown as a red dashed line) intersects the original geometric boundary. (b) Temperature distribution in the geometric domain in (a). (c) Adding another boundary piece to the geometric model in (a) changes the way how the closing lines are created. Now they do not intersect with the original geometric boundaries. (d) Temperature distribution in the geometric domain in (c). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

geometric models are caused by the boundary conditions assumed for the closing boundaries. In our numerical examples we used default natural boundary conditions which correspond to heat insulation in the case of heat transfer problems, and to a traction free boundary in the case of structural analysis problems. Since analysis of the numerical distortions in the solution caused by the geometric imperfections is outside of the scope of our paper, we will investigate these issues in our future work. We will also work on extension and generalization of the numerical integration algorithms into 3D. A straightforward extension of the proposed technique could be based on representation of a 3D geometric model by a set of parallel 2D cross-sections that are allocated according to the Gauss integration rule in one of the coordinate directions (for example, in “Z” direction). In this case a 3D volume integral appears as a weighted sum of 2D integrals in the cross-sections.

Acknowledgments

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